
A MEMORY THAT ONLY LOOKS LIKE ONE: THE RECURRENT STATE OF A HYBRID VISION-LANGUAGE MODEL RELAYS THE PICTURE RATHER THAN STORING IT

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ABSTRACT

A hybrid vision-language model carries two kinds of memory. Its attention layers keep a growing cache: every token the model reads, including every patch of an image, is stored as a key-value entry that later tokens can look up. Its recurrent layers keep a single state of fixed size, and every new token overwrites part of that state. Prior work proved that a fixed-size state must eventually lose information. But it does not tell us *when* a given model forgets, *what* it keeps while forgetting, or *what breaks* downstream. We answer these three questions for Zamba2-VL-7B, a model in which all 81 decoder layers carry a Mamba2 recurrent mixer and only 13 also carry attention. Put simply: while the model talks about other things, where does the picture live?

First, what does the recurrent state hold just after it reads an image? It holds the gist, not the details. A linear probe reads *which* object was in the picture at 92.5% accuracy from the best single layer, against a 12.5% chance rate. The same probe can barely read *how many* objects there were, or *where* they sat.

Second, how fast should that memory fade? For this architecture the answer can be computed from the weights alone, with nothing fitted. Mamba2 multiplies its state by $\exp(\Delta_t A)$ at every token, so retention after k tokens is $\exp(A \sum \Delta_t)$, and each head has a half-life of $\ln 2 / (|A| \bar{\Delta})$ tokens. The median half-life is 6.6 tokens in the 7B model, and 2.9 and 2.6 in its two smaller siblings. The weights say the recurrent trace of the picture should be gone within a few tokens. (This calculation predicts which heads depend on attention, but it does not predict the measured decodability curve across scales; we report that failure.)

Third, and this is the central result: the memory a probe appears to find at longer distances is mostly not memory. We split heads into fast-decaying and slow-decaying groups while holding write strength fixed; the same gate controls both, so a naive split confounds them. The ordering comes out exactly as decay predicts, with slow heads reading better at every distance. But the amounts are impossible. Heads with a 2.3-token half-life still decode the object at 26.6% sixty-four tokens after the image, where their own parameters retain 4×10^{-9} of what was written. Something must be re-supplying them. To find out what, we delete the image’s key-value entries from all 13 attention layers immediately after the image, without touching any recurrent state. The apparent memory collapses: the fast third of heads falls from 51.6% to 7.8% at thirty-two tokens, below chance. At matched write strength the deletion costs fast heads 2.4 times more than slow heads, exactly the pattern a relay predicts. The recurrent channel is passing along what attention still holds; it is not storing the picture.

A direct causal test agrees. We swapped the entire recurrent memory, all 81 states, between two runs that saw different images. Nothing changed: 100 of 100 answers still followed the run’s own image, with every swap verified. Swapping seven of the thirteen attention caches flipped 94 of 100 answers to the other image. Finally, the practical consequence for cache pruning: on this model, keeping a quarter of the visual key-value entries is safe at every distance we tested, but below that the cost

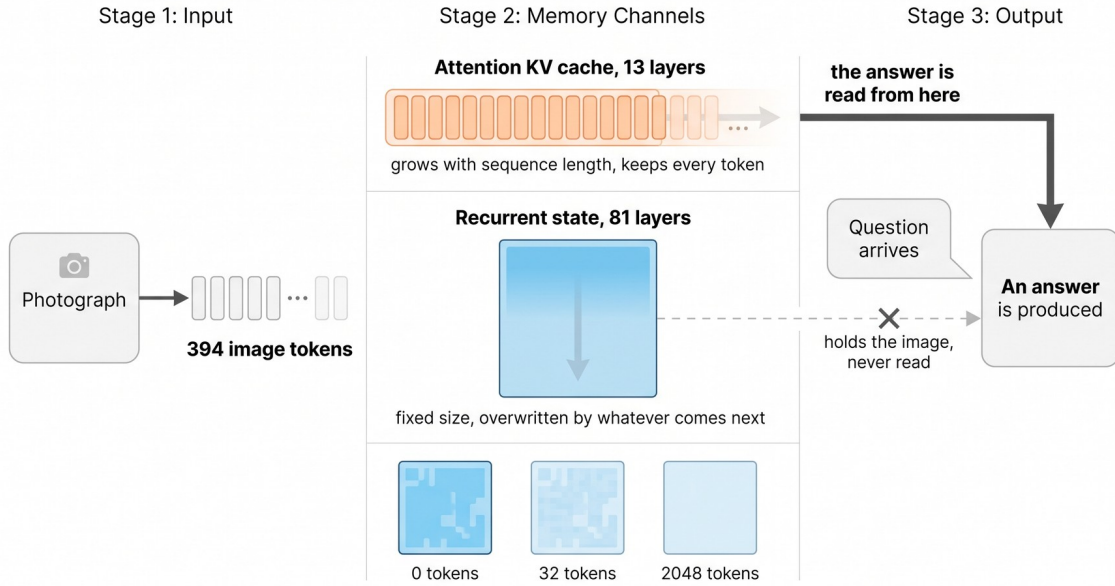


Figure 1: The two memory channels of a hybrid vision-language model and the question this paper asks of them.

grows with distance. Removing every visual entry costs 34 accuracy points next to the image and 59 points thirty-two tokens later.

1 What Was Known Before, and What Is New Here

Linear-attention and state-space models replace the growing key-value cache of a transformer with a state of fixed size [1, 2, 3, 4, 5, 6]. A fixed state cannot grow with the context. For this reason, most deployed systems keep a few attention layers alongside the recurrent ones, and these hybrid models are now common [7, 8, 9, 10, 11]. This paper asks a simple question about such a model: what is the recurrent half actually holding?

We want to be clear about what is already known. Prior work established that a fixed-size recurrent state must eventually lose information from context; that premise is not ours. Jelassi et al. [12] prove that transformers can copy exponentially long strings while state-space models are, in their words, “fundamentally limited by their fixed-size latent state”. They also show empirically that pretrained transformers “dramatically outperform state space models at copying and retrieving information from context”. Other results reach the same conclusion from other directions. State-space models are limited to TC^0 and cannot reliably track state [13]. In-context retrieval is the specific skill that separates recurrent models from transformers [14]. Most of the quality gap between efficient architectures and attention comes from recall [15, 16]. Our work builds on these results; it does not rediscover them.

Table 1 states what that literature establishes and what this paper adds.

2 Significance

Why does this matter? Hybrid models are being adopted because a fixed-size state promises constant-memory inference. Anyone building or deploying one must then decide how much to trust the recurrent channel: can it be relied on to carry information across a long prompt? Today that is a judgement call. This paper turns it into a calculation. Given a checkpoint’s decay weight A and its gate statistics, you can compute how many tokens the recurrent state retains what it wrote, before running any experiment. For Zamba2-VL-7B the answer is single digits. That is far shorter than one would guess from the size of the state, which is 458,752 numbers per layer.

There is also a concrete consequence for a popular optimization. A large body of work speeds up vision-language models by discarding visual tokens or evicting their key-value entries. The stated premise is that those entries are largely redundant [24, 25, 26]; a parallel line evicts key-value entries in language models on similar grounds [27,

Established by prior work	New here
A fixed-size state must lose information from context, with a capacity proof and behavioural evidence [12, 13, 14].	<i>When.</i> A retention horizon in tokens, computed from the model’s own A and Δ with nothing fitted, at three scales.
Recurrent models underperform at copying and retrieval as a class [15, 16].	<i>What survives.</i> The failure is content-selective: object identity is highly decodable at the same instant that count and position are near chance.
Architectures are compared to one another from the outside.	<i>Causal localisation inside one model,</i> and the finding that the recurrent channel relays rather than stores, which makes probing it at distance a measurement of the attention channel.
Visual tokens and key-value entries carry substantial redundancy that can be pruned [24, 25, 26].	<i>What underwrites that premise,</i> and that its safety margin shrinks with distance from the image on a hybrid.

Table 1: Delineation of prior work and contribution.

28, 29]. These methods were developed and validated on dense transformers, where they work. We find that they mostly survive the move to this hybrid: keeping a quarter of the visual entries holds accuracy at every distance we tested. What does not survive is the safety margin. The redundancy exists because the recurrent channel is relaying the picture, and it only relays for a few tokens. As a result, the cost of aggressive pruning roughly doubles between a prompt that asks about the image immediately and one that asks thirty-two tokens later. A pruning method tuned on short prompts will therefore look safer than it is. That is the kind of failure a standard benchmark is least likely to catch.

3 The Two Memories a Hybrid Carries

A transformer remembers by accumulation. Every token it reads leaves a key and a value in a cache, and every later token may attend to all of them. Nothing is lost and nothing decays. The cost is that the cache grows with the context. This growth is why serving long inputs is expensive.

A linear-attention or state-space layer remembers by summary. It keeps one state of fixed size and updates it at every token, so its memory cost does not grow at all. The cost appears somewhere else. The state is the same size after ten tokens as after ten thousand, so everything it holds must share that space. What a later token writes displaces what an earlier one wrote.

A hybrid model runs both at once. In Zamba2-VL-7B, all 81 decoder layers carry a Mamba2 mixer with such a state, and 13 of them also carry an attention block with an ordinary growing cache. Figure 2 draws the situation at the moment the model is asked a question about an image it read d tokens earlier. The attention channel still holds the image exactly as it was read, because nothing has removed those entries. The recurrent channel holds whatever survives after d rounds of overwriting.

Notice that the two channels give conflicting hints about where the picture should be. The recurrent channel is much larger: 81 states of 458,752 numbers each, 37,158,912 numbers in total, against 13 caches holding a few hundred entries apiece. If capacity decided the question, the picture would be in the recurrent channel. But the recurrent channel is also the one that overwrites itself. The architecture alone cannot tell us which channel the model actually answers from. The rest of this paper settles the question by experiment.

4 Model and Instruments

4.1 The model

Zamba2-VL-7B [35] is the vision-language member of the Zamba family [8], which pairs a Mamba2 backbone [5] with a shared attention module, and it takes its vision encoder from Qwen2.5-VL [31]. It is Apache-2.0 licensed and has 81 decoder layers. Every one carries a Mamba2 mixer, and 13 of them additionally carry a shared attention block, at layer indices 6, 11, 17, 23, 29, 35, 41, 47, 53, 59, 65, 71 and 77. The recurrent channel therefore spans all 81 layers and the attention channel spans 13. Each layer’s recurrent state is a tensor of shape [112, 64, 64], which is 458,752 numbers. One image occupies between 345 and 415 tokens depending on its aspect ratio.

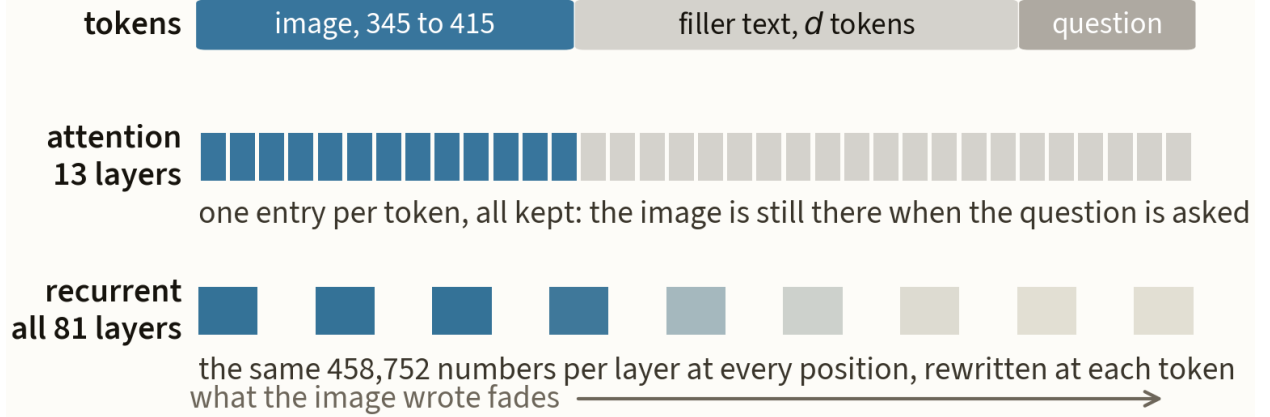


Figure 2: The two memories at the moment the question arrives. **Top:** the token stream, an image of 345 to 415 tokens followed by d tokens of filler and then the question. **Middle:** the attention channel appends one key-value entry per token and keeps all of them, so the image is still there when the question is asked. **Bottom:** the recurrent channel is the same 458,752 numbers per layer at every position, rewritten at every token, so what the image wrote fades as the filler streams past. The question this paper answers is whether it fades on the schedule the weights dictate. Schematic; the quantities are those of Section 4.

4.2 Reading the state

The Mamba2 recurrence is

$$S_t = S_{t-1} \exp(\Delta_t A) + \Delta_t x_t B_t^\top, \quad y_t = S_t C_t, \quad (1)$$

so writing uses (x, B) , reading uses C , and retention of anything already present is governed by A and Δ alone.

Let us read that update in words, because everything in this paper follows from it. At each token, two things happen to the state. First, it is multiplied by $\exp(\Delta_t A)$, a number strictly between zero and one. Second, the term $\Delta_t x_t B_t^\top$ is added to it. The multiplication is the only operation that ever removes information: it shrinks everything the state already held, uniformly, by a factor the model sets through the stored weight A and the input-dependent gate Δ_t . The addition is the only operation that ever writes. Reading the state, through C_t , changes nothing.

In other words, each head is a leaky accumulator, and its leak rate can be computed in advance. Multiplying by $\exp(\Delta A)$ once per token for k tokens leaves $\exp(A \sum_t \Delta_t)$ of whatever was there. Half of the content is gone after $\ln 2 / (|A| \bar{\Delta})$ tokens, so we call this quantity the head’s half-life. Notice what the expression does *not* contain: nothing about the image, the task, or the probe. It is one stored weight and one gate statistic, and both can be read off a checkpoint before any experiment is run. Section 7 tests this prediction.

We verified each of these against the installed model source rather than assuming the canonical form. In particular $A = -\exp(A_{\log})$ is a per-head scalar, the gate is $\Delta = \text{softplus}(\Delta_{\text{raw}} + b)$ clamped below at 0.001 with no upper clamp, and the decay factor is $\exp(\Delta_t A)$ elementwise.

To measure what the state holds, we use a linear probe. A linear probe is the simplest possible reader: a classifier that computes a weighted sum of the state’s numbers and predicts the trial’s answer from that sum alone, with no other computation. We train it on some trials and test it on held-out ones. If even this simple reader recovers the answer, the answer is present in the state in an easily readable form. Probing a representation this way is standard [17], and its hazards are well known. A probe can succeed because the classifier is powerful, not because the model uses the information. This is why probing papers check selectivity against control tasks [18] and read their results with care [19]. We therefore never rely on a probe alone. Every probe result in this paper is paired with an intervention on the model, in the tradition of causal mediation analysis [20] and activation patching [21, 22, 23], and we follow that literature’s recommendation to report what each intervention does to the model’s own behaviour, not only to the probe’s readout [23].

One practical detail: 458,752 numbers per layer is too wide to probe directly, so each layer’s state is first passed through a fixed Gaussian random projection down to 512 features. By the Johnson-Lindenstrauss property, such a projection preserves linear separability up to a controlled distortion. A linear probe on the projected features is therefore, to that tolerance, a linear probe on the state itself. The projection columns are drawn independently, so the first k columns form a valid k -feature projection on their own.

Probes are ridge classifiers evaluated by block-grouped k -fold cross-validation, with the ridge penalty chosen on inner training folds only. One rule matters enough to state on its own: every probe is fitted and evaluated *within a single condition*. A probe reported for the evicted condition is trained only on evicted trials and tested on held-out evicted trials. It is never trained on intact trials and applied to evicted ones. Why does this matter? If we had transferred a probe across conditions, a below-chance readout would be the expected signature of a distribution shift between conditions, and it would tell us nothing about what information is present. Because each condition supplies its own training and test folds, a drop under eviction means one thing: less linearly decodable information in that condition. The below-chance cells reported later are within sampling noise at these sample sizes. A reading of 7.8% is 5 correct out of 64 ($P(X \leq 5) = 0.17$ under the null), and 4.7% is 3 of 64 ($P = 0.034$), so across twenty cells roughly three readings below chance are expected by chance alone.

4.3 Splicing the channels

The splice experiment runs the model twice, on two different images, and then physically exchanges one memory channel between the two runs while leaving the other channel untouched. The run that keeps its own question is called the host; the run that donates its memory is called the donor. After the exchange, the host answers its question, and we record whose image the answer describes. Swapping the recurrent channel replaces all 81 states and convolution buffers. Swapping the attention channel replaces the 13 key and value caches. The host’s cache is restored exactly afterwards, so one prefill supports many conditions.

4.4 The three experiments in one picture

Three protocols recur throughout the paper, and Figure 3 shows what each one does to a run. The first is the probe: read the recurrent state just before the question, and measure what is linearly present at a chosen distance. The second is the eviction: delete the image’s key-value entries from all 13 attention layers at the moment the image finishes, before a single filler token is processed, and then probe the state exactly as before. At the instant of the deletion, the recurrent state is identical in both runs. Any difference that appears later must therefore come from the attention channel. The third is the splice described above, which measures which channel the model’s own answer follows.

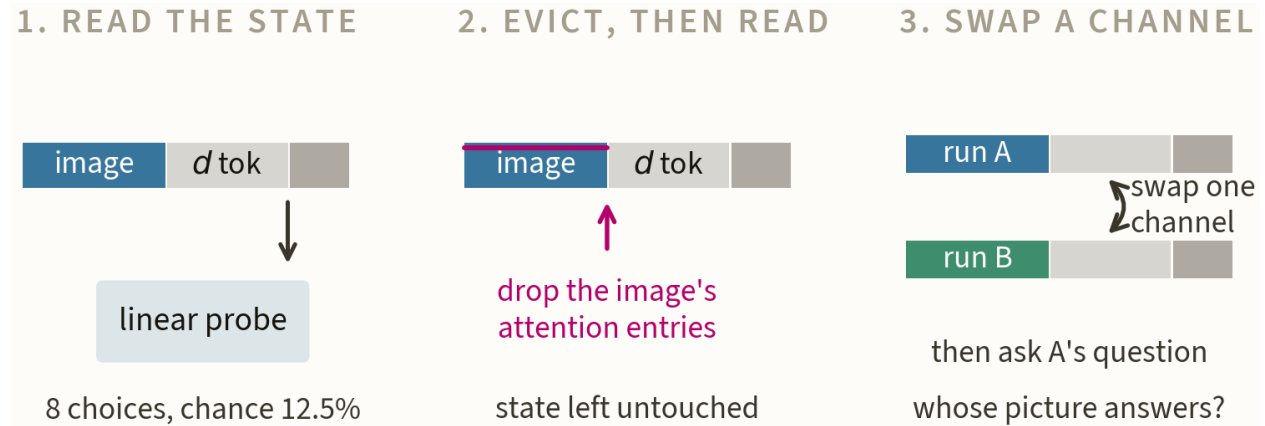


Figure 3: The three protocols. **1:** a linear probe reads the recurrent state at the position just before the question; the task is eight-way forced choice, so chance is exactly 12.5%. **2:** the image’s attention entries are deleted at the moment the image ends, before any filler token is processed, leaving the recurrent state untouched and identical to the intact run at that instant. **3:** one channel is exchanged between two runs that saw different images, and the host run’s question is then asked. Schematic.

5 Materials

Trials are built from COCO val2017 instance annotations [34], the corpus underlying most visual instruction tuning [30]. There are three question families: which object is present, how many of a named object are present, and where in the picture a named object is. Every question is eight-way forced choice, so the chance rate is exactly 12.5%. Trials are constructed in blocks of eight. All eight trials in a block share one identical candidate list, and each candidate is the correct answer exactly once within the block. This construction is what makes the pre-image control meaningful. Because the candidate set is fixed and each answer appears equally often, a classifier that has not seen the image has nothing to exploit: no candidate is more likely than any other, so it cannot beat chance.

Images are fetched at native geometry for the decay experiments. For the splice they are letterboxed onto a common square canvas, because two runs on images of different shape emit different numbers of vision tokens and therefore different-length attention caches, which cannot be exchanged. Letterboxing rather than square resizing matters: in a pilot, a plain square resize deformed the content enough to halve identity accuracy, so every splice image is letterboxed.

6 Result 1: The State Records Gist, Not Particulars

We first probe the state at distance zero, the instant the image has just been read. The state is highly informative about which object is present, and close to uninformative about how many objects or where they are. The best single layer reads identity at 92.5% against the 12.5% chance rate. A probe given all 81 layers at once reads identity at 83.1% ($p = 6 \times 10^{-92}$, $n = 160$), count at only 19.4% ($p = 0.009$), and position at only 17.5% ($p = 0.041$). Note that the all-layer figure is pulled down by overfitting, because it fits 41,472 features to 120 trials. We therefore use it only to compare the three properties against each other, never as a measure of how much the state holds.

In fact these figures understate what the state carries. The full-state feature vector has $81 \times 512 = 41,472$ dimensions, far more than the number of trials, and a probe on a single layer does substantially better. Layer 64 alone reads identity at 92.5%. The last 21 layers (60 to 80) average 85.2%, against 59.0% for the middle third of the stack (layers 27 to 53). On the identical trials, the full-state probe reads 71.7%. Two conclusions follow. The gist is stronger than the aggregate number suggests, and it is concentrated in the late layers, as the right panel of Figure 4 shows.

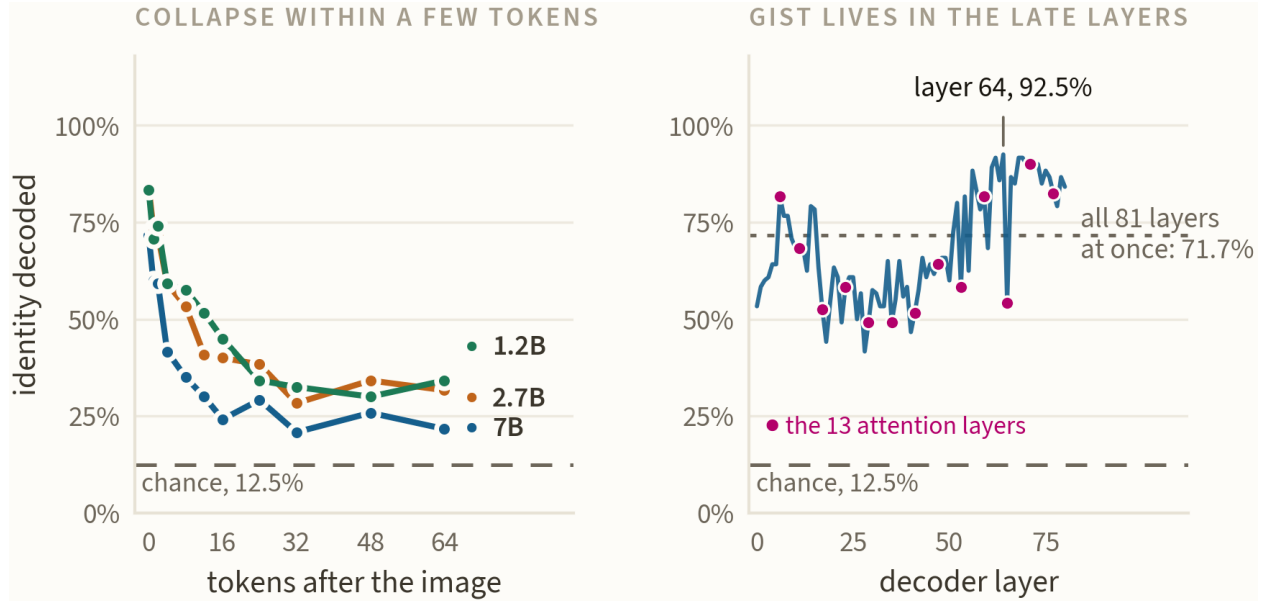


Figure 4: **Left:** identity decoded from the recurrent state against distance from the image, at three model scales, 120 trials per point. The pre-image control reads exactly 12.5% at every scale. **Right:** identity decoded from each layer’s state alone at the instant of writing, with red marking the 13 layers that also carry attention. A probe given all 81 layers at once does worse than the best single layer, because 41,472 features on 120 trials is heavily overparametrised.

The recurrent state is not a compressed copy of the image. It is a selective summary that keeps object identity and discards count and position, and that selection is visible at the instant of writing rather than emerging over distance.

7 Result 2: The Forgetting Horizon, Predicted From the Weights

7.1 The prediction

Retention of whatever the state held, after k further tokens, is

$$R(k) = \exp\left(A \sum_{t=1}^k \Delta_t\right), \quad \text{half-life} = \frac{\ln 2}{|A| \overline{\Delta}}. \quad (2)$$

A is a stored weight. $\bar{\Delta}$ is measured from the gate channels of the input projection, over the tokens that follow the image, on the same prompts the decay experiment uses. Nothing here is fitted to a decay curve.

Across all $81 \times 112 = 9,072$ heads, the predicted median half-life is 6.6 tokens, with an interquartile range of 2.8 to 17.8. 85.0% of heads fall below 32 tokens, and only 0.08% exceed 512. Figure 5 draws the full distribution: a dense cluster of heads in the single digits, plus a thin tail of slow heads reaching past a hundred tokens. Keep that tail in mind. When a probe still reads something at long distance, the most natural explanation is that this slow minority carries it. Section 8 tests that explanation directly and finds it insufficient: the slow heads do keep more, but nowhere near enough to explain the readings.

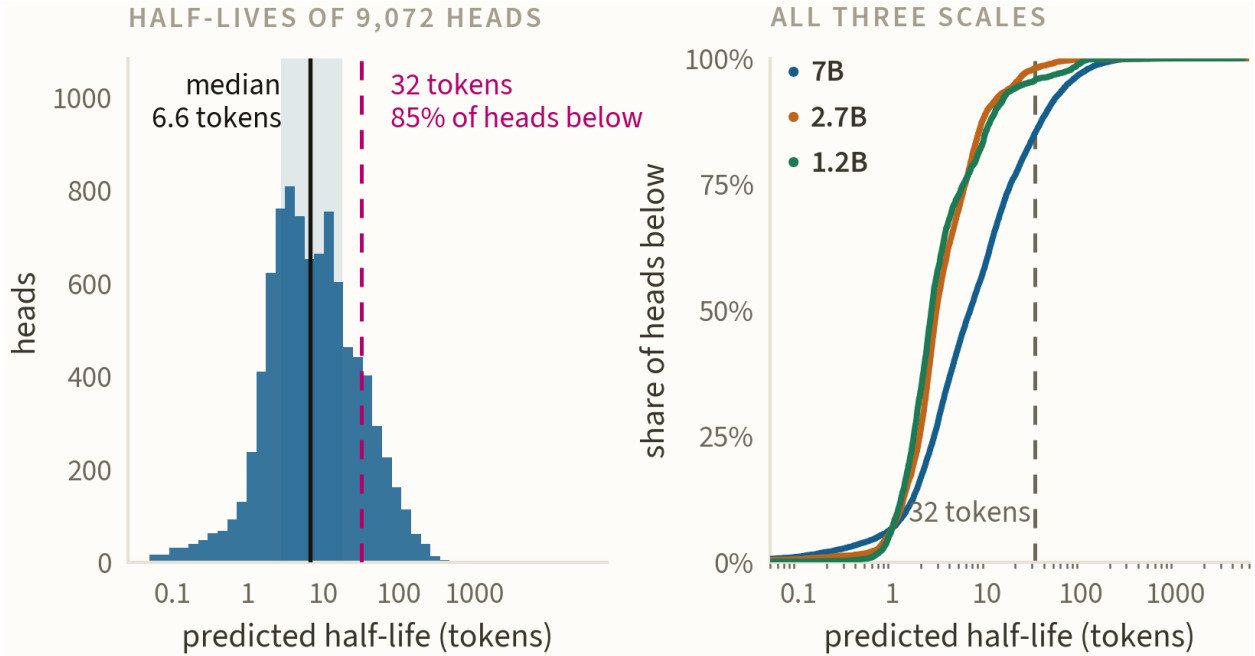


Figure 5: **Left:** predicted half-life for each of the 9,072 heads of Zamba2-VL-7B, from that head’s own A and its mean gate over the filler. The median is 6.6 tokens and 85% of heads fall below 32. **Right:** the same quantity at three model scales, as the share of heads whose half-life falls below a given value. The distributions differ in detail but not in kind; at every scale the bulk of the model forgets within tens of tokens.

Measured over eight different photographs the median moves only between 6.44 and 6.86 tokens, so the horizon is a property of the model rather than of the stimulus.

7.2 The measurement

Does the measured decay match this prediction? Our original collection measured only distances 0 and 32, and 32 tokens is already far past the region where the prediction changes quickly. Table 2 therefore shows a finer sweep, on 120 identity trials at each of eleven distances.

The measured signal halves at 4.0 tokens, which falls inside the predicted interquartile range of 2.8 to 17.8. Normalised to its value at distance zero, measured decodability correlates with predicted mean retention at $r = 0.973$ across the eleven distances, with no free parameters. Figure 6 shows both halves of this comparison. The left panel is the prediction on its own: three retention curves, one per stratum, each produced by multiplying by $\exp(-|A|\bar{\Delta})$ once per token and doing nothing else. The right panel compares that prediction with what the probe actually read, distance by distance.

Beyond 16 tokens the measured curve sits above the predicted median and between the median and the mean, which is what the heavy tail of slow heads predicts: the residual signal at long distance is carried by the minority of heads with long half-lives.

Throughout this collapse the model’s own accuracy never moves. It answers correctly on 94 to 98% of trials at every distance, including those where the state has been emptied.

tokens	probe accuracy	p	behaviour	predicted retention (median)
0	71.7%	2×10^{-50}	94.2%	1.000
1	60.0%	2×10^{-34}	96.7%	0.901
2	59.2%	2×10^{-33}	97.5%	0.812
4	41.7%	1×10^{-15}	96.7%	0.659
8	35.0%	2×10^{-10}	97.5%	0.434
12	30.0%	3×10^{-7}	96.7%	0.286
16	24.2%	3×10^{-4}	97.5%	0.188
24	29.2%	1×10^{-6}	95.8%	0.082
32	20.8%	7×10^{-3}	95.0%	0.036
48	25.8%	6×10^{-5}	97.5%	0.007
64	21.7%	3×10^{-3}	98.3%	0.001

Table 2: Decodability of object identity against distance from the image, on 120 trials at every distance, with the retention predicted from the weights alongside. Chance is 12.5%, and the pre-image control reads exactly 12.5%. Notice the shape: the reading falls fast, as the weights predict, and then stops well short of chance. At 64 tokens the predicted retention is 0.001 of the trace, yet the probe still reads 21.7%. That tension is resolved in Section 8. Notice also the behaviour column: the model itself answers correctly at every distance throughout.

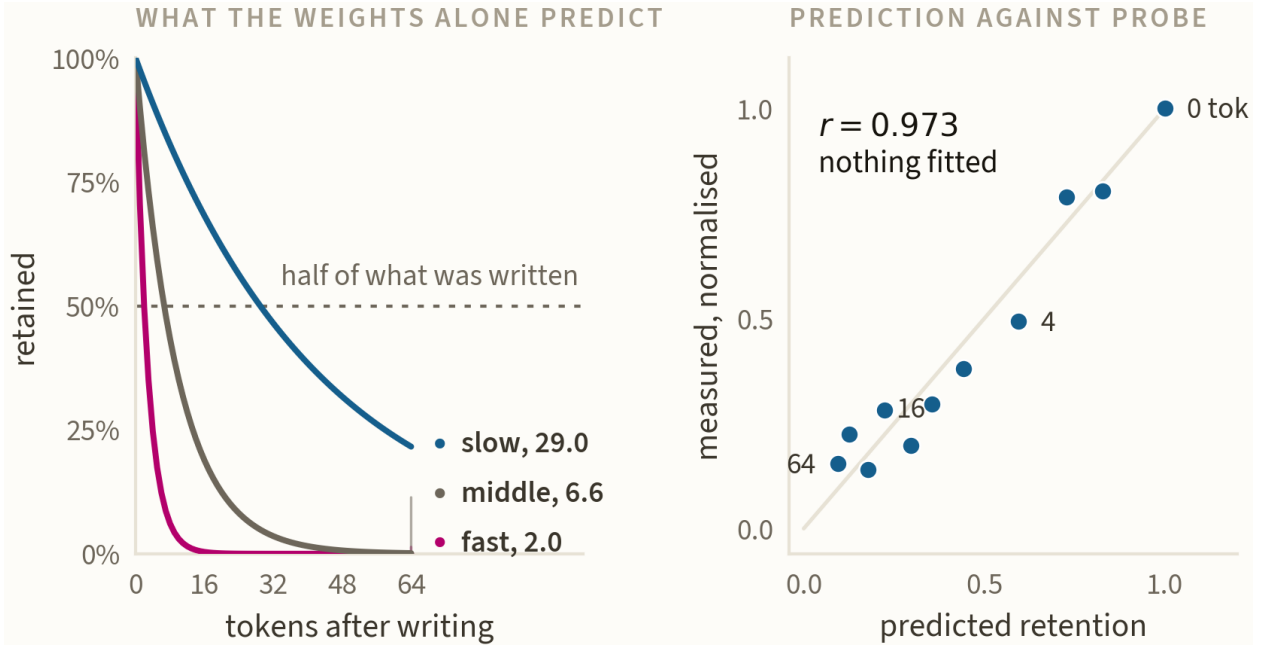


Figure 6: **Left:** what the weights alone predict. Each curve is $\exp(-|A|\bar{\Delta}k)$ at the median rate of one third of the heads, labelled with that third’s median half-life. Nothing is fitted. **Right:** predicted retention against measured decodability, normalised to distance zero and labelled by distance in tokens. The agreement on this model is $r = 0.973$. Section 10 reports that it does not replicate across scales, and Section 8 explains why it cannot.

7.3 The within-model test, with write amplitude held fixed

A curve comparison alone cannot rule out coincidence. So we ran a sharper test. Every head has its own predicted half-life, so we can sort them. We call a head *fast* if its half-life is short, around two tokens, meaning its state should empty almost immediately, and *slow* if its half-life is long, meaning it should hold information longest. We then probe the fast group and the slow group separately, using the same projection basis so the groups are comparable to each other and to the unstratified probe. If the state is genuinely retaining, decay makes a clear prediction: the fast heads should lose the image first, and whatever signal survives at long distance should be carried by the slow heads.

Before giving the result, we must explain a confound, because it changed our own conclusions. The half-life is $\ln 2 / (|A|\bar{\Delta})$, and the gate $\bar{\Delta}$ appears in the write term as well. The gate therefore does double duty: it sets how fast a head forgets, and it also sets how strongly the head writes, by about a factor of seven between the fastest and slowest

thirds. Why does writing harder matter? Because a head that writes with larger numbers leaves a stronger signal in its state, and a stronger signal is easier for a linear probe to read, regardless of where the information came from or how old it is. Sorting heads by half-life therefore also, accidentally, sorts them by signal strength. The naive split indeed produces a striking reversal, with the fast third decoding identity at 61.7% after sixty-four tokens while the slow third falls to 16.7%, the opposite of the decay ordering. That reversal has nothing to do with memory. It is the write strength.

The fix is to compare only heads that write equally hard. We keep the 3,628 heads whose mean gate falls in the middle forty percent of its distribution, so write strengths are nearly matched, and split those by the decay weight $|A|$ alone. This gives two strata of 907 heads each. Their predicted median half-lives still differ sevenfold, 2.3 against 16.4 tokens, but their write gains are now nearly equal (median $\bar{\Delta}$ of 0.074 against 0.066). The only meaningful difference between the two groups is how fast they forget.

With the gate controlled, the reversal disappears (Table 3, left panel of Figure 7). The slow stratum decodes better at all seven distances measured, and it keeps a larger fraction of its distance-zero signal at sixty-four tokens, 0.28 against 0.20. The ordering that the decay parameters predict is real; amplitude had been masking it.

stratum	predicted $t_{1/2}$	$d=0$	$d=16$	$d=32$	$d=64$	signal kept at $d=64$
fast	2.3	81.8%	50.0%	35.9%	26.6%	0.20
slow	16.4	91.7%	61.5%	46.9%	34.4%	0.28

Table 3: Identity decodability by gain-matched stratum, $n = 192$ per cell, chance 12.5%. “Signal kept” is the above-chance excess at $d=64$ as a fraction of the excess at $d=0$. The slow stratum keeps more at every distance, which is the decay ordering; both strata keep far more than their own decay parameters permit.

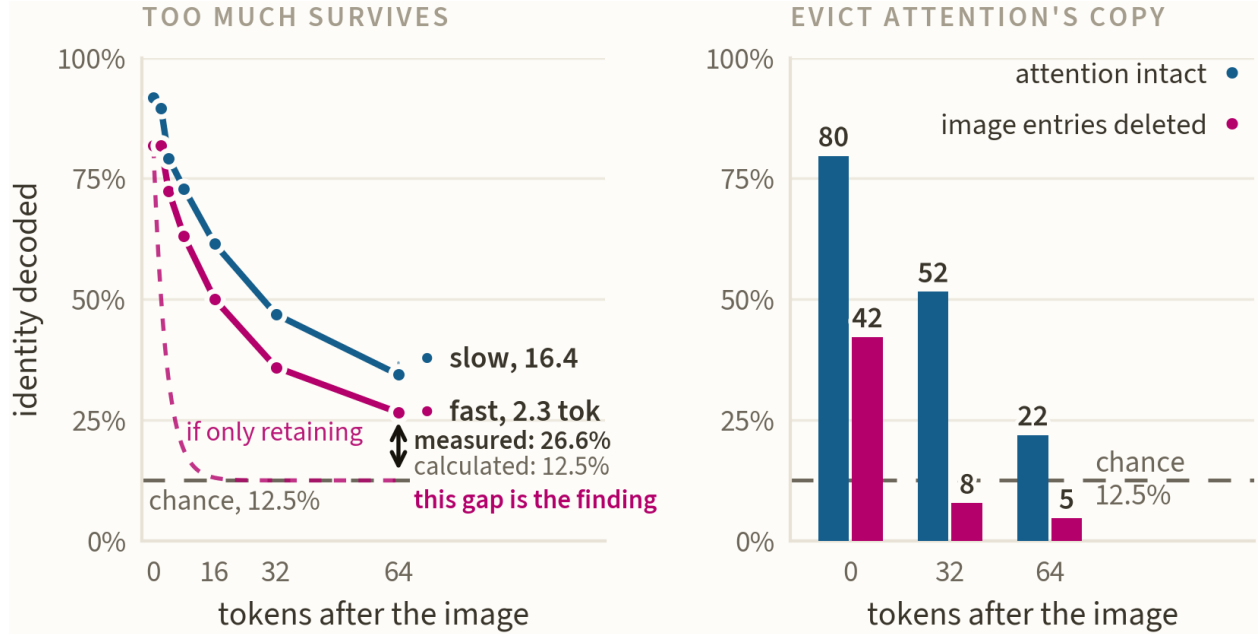


Figure 7: Left: how to read this panel. The two solid curves are measurements: identity decoded from the slow stratum and from the fast stratum at each distance, with write gain matched and $n = 192$ per point. The dashed curve is not a measurement. It is a calculation: where the fast stratum's reading would fall if it came only from what its own 2.3-token half-life retains. That calculated curve reaches chance within ten tokens; the measured one ends at 26.6% at $d=64$. The solid curves being ordered slow-above-fast is what decay predicts; the gap between the two fast curves is what decay cannot explain. **Right:** the same recurrent state probed with and without the attention channel's copy of the image, fast tercile, $n = 64$ per bar. The eviction happens immediately after the image is read and never touches the recurrent state, which is identical in both runs at that instant.

But now look at the amounts, because they are the finding. A stratum with a predicted half-life of 2.3 tokens retains $2^{-64/2.3} \approx 4 \times 10^{-9}$ of anything it wrote sixty-four tokens ago. Yet it reads 26.6% there, twice the chance rate ($p = 1 \times 10^{-7}$, $n = 192$), and the measured half-life of its readout is 19.3 tokens, eight times its prediction. The slow stratum overshoots in the same direction: predicted 16.4 tokens, measured 26.3, with four times more signal surviving

at $d=64$ than its parameters allow. In short, the strata are ordered exactly as decay predicts, and both hold far more signal than decay permits.

One caveat applies to any global split of this kind. Half-life varies across the stack by almost two orders of magnitude (Figure 8), from a median of 1.7 tokens at the fastest layer to 84.7 at the slowest. A global split by a decay quantity is therefore also a partial split by depth, and depth matters for decoding. This affects the comparison *between* strata. It does not affect the comparison of each stratum with its own prediction: whatever layers the fast stratum draws from, its own decay parameters bound what it can retain, and it reads far above that bound.

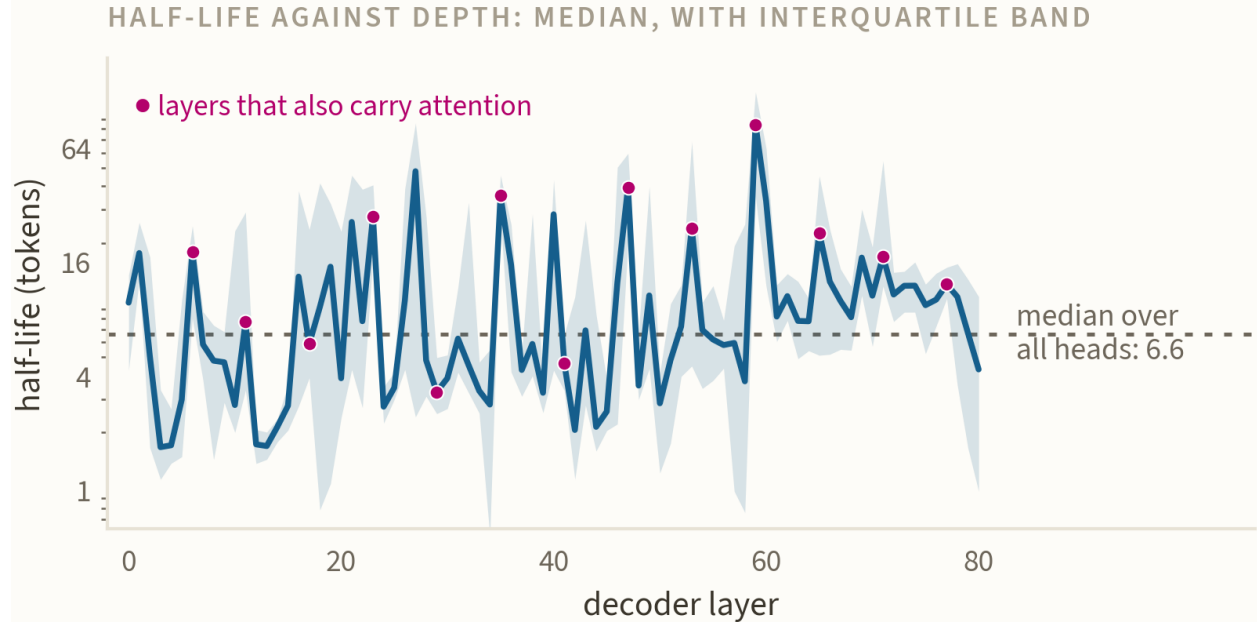


Figure 8: Predicted half-life against depth for Zamba2-VL-7B: the median over the 112 heads of each layer, with the interquartile band. The spread within a single layer is often wider than the difference between layers, which is what makes a global split by half-life only a partial split by depth. Red marks the 13 layers that also carry attention.

With write amplitude held fixed, the strata are ordered exactly as decay predicts, and both hold far more signal than decay permits. A head with a two-token half-life cannot be holding something written sixty-four tokens earlier: by then its state consists almost entirely of recent writes. If such heads still decode object identity at twice chance, the identity information in their state arrived recently. Something else in the model must be supplying it.

8 Result 3: The Recurrent Channel Is a Relay, Not a Store

What could be supplying it? Only one component of the model still holds the image at distance: the attention channel, whose key-value entries do not decay. We tested this directly. Each trial was run twice, identically, except that in one run we deleted the image’s key-value entries from all 13 attention layers immediately after the image was read, *before a single filler token was processed*. The deletion does not touch the recurrent state. At that instant, the recurrent state is identical in both runs.

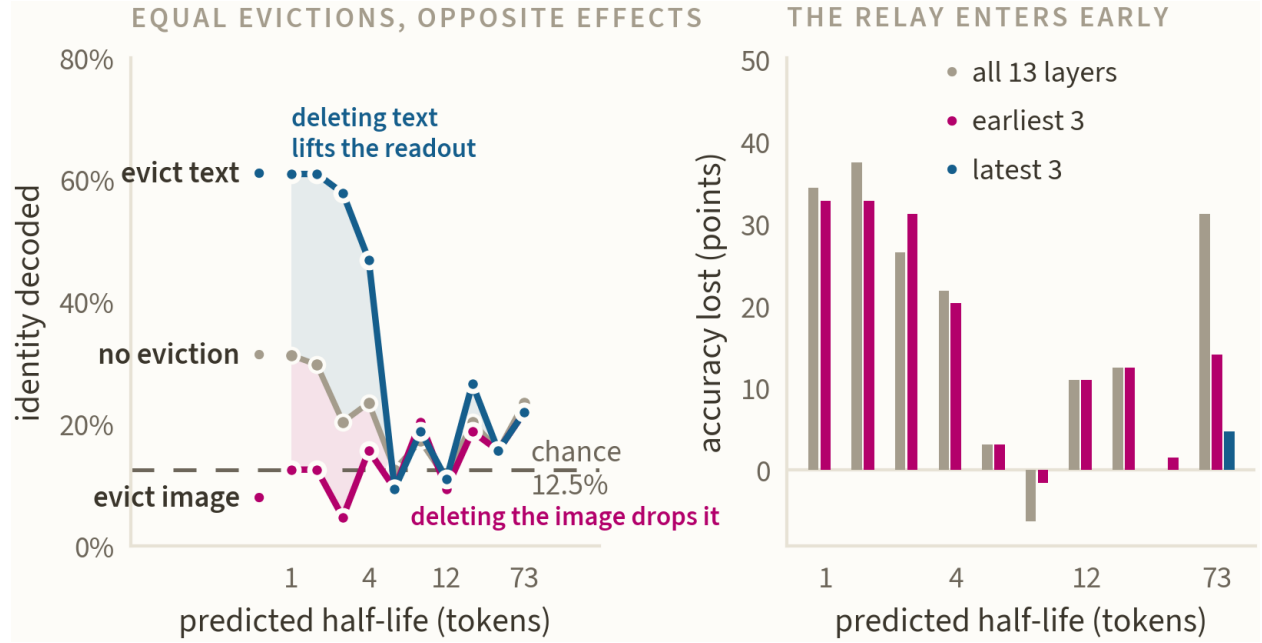
The two hypotheses now make opposite predictions. If the recurrent state is remembering the image on its own, deleting attention’s copy should not change what a probe reads from the state. If the state is being refreshed from attention, the deletion should collapse decodability at distance, and it should hurt the fast heads most, because they depend most on being refreshed.

8.1 Controls: the eviction removes the image, not merely positions

There is a fair objection to the eviction experiment: the deletion changes more than the image’s presence. Removing the visual entries shortens every attention cache by roughly 400 positions, and it changes the softmax normalisation for every token that follows. Perhaps the readout collapsed because we damaged the machinery, not because we removed a memory. Here is the experiment that settles it. Run the deletion twice, on the same trials. In one version, delete

stratum	distance	visual KV intact	visual KV evicted	change
fast	0	79.7%	42.2%	-37.5
fast	32	51.6%	7.8%	-43.7
fast	64	21.9%	4.7%	-17.2
middle	0	62.5%	43.8%	-18.8
slow	0	62.5%	54.7%	-7.8
all	0	59.4%	59.4%	0.0
all	32	6.2%	6.2%	0.0

Table 4: Identity decodability from the recurrent state with and without the attention channel’s copy of the image, $n = 64$ per cell. Chance is 12.5%. The fast heads’ apparent memory at 32 tokens is entirely dependent on the attention channel: removing it drops them from 51.6% to below chance.



the image’s entries, exactly as before. In the other, delete the same *number* of positions from the same caches, but delete text instead of the image. Both versions do identical mechanical damage: the same number of entries gone, the same shorter caches, the same disturbance to attention. So if damage were the explanation, both versions should push the reading down. We ran this control on ten strata formed by deciles of predicted half-life, with 64 trials per cell (Table 5).

That is not what happens. The two deletions move the readout in opposite directions, shown in the left panel of Figure 9. At 32 tokens, deleting the image costs the fastest decile 18.8 points, while deleting an equal number of the most recent pre-image text positions *raises* it by 29.7 points. Averaged over the ten strata, the two effects are -6.4 and $+12.5$ points. Deleting text plausibly helps because the text was competing with the image for the residual stream, and removing it leaves the image easier to relay. The key point is the signs. Identical mechanical damage moves the reading in opposite directions, and damage cannot push a reading up. The same dissociation holds on the gain-matched strata at three times the trial count ($n = 192$ per cell): deleting the image beside it costs the fast stratum 22.4 points while deleting matched text *raises* it 13.5, and at thirty-two tokens text deletion raises the two strata by 31 and 32 points while image deletion moves them by $+1$ and -7 .

decile	pred. $t_{1/2}$	$d = 0$			$d = 32$	
		intact	evict image	evict text	intact	evict image
1	0.70	56.2%	18.8%	78.1%	31.2%	12.5%
2	1.84	56.2%	17.2%	79.7%	29.7%	12.5%
3	2.75	53.1%	18.8%	73.4%	20.3%	4.7%
4	3.79	46.9%	21.9%	71.9%	23.4%	15.6%
5	5.38	31.2%	26.6%	42.2%	12.5%	9.4%
6	8.15	35.9%	40.6%	51.6%	17.2%	20.3%
7	11.8	37.5%	25.0%	54.7%	10.9%	9.4%
8	17.8	45.3%	29.7%	57.8%	20.3%	18.8%
9	32.0	28.1%	26.6%	32.8%	15.6%	15.6%
10	73.3	51.6%	29.7%	46.9%	23.4%	21.9%

Table 5: The matched-eviction control by decile of predicted half-life, $n = 64$ per cell, chance 12.5%. “Evict image” and “evict text” delete the same number of positions from the same 13 attention caches. Single cells carry about 6 points of binomial noise at $n = 64$, and two of the twenty image-eviction cells fall slightly the wrong way; the trend rather than any one cell is the result. The drop under image eviction correlates with the decay rate $1/t_{1/2}$ at $r = 0.647$ (95% CI [0.03, 0.91]) at $d=0$ and $r = 0.800$ ([0.34, 0.95]) at $d=32$. With ten strata these intervals are wide, and the lower bound at $d=0$ is close to zero.

The same collection also tells us where the relayed information enters the model, in the right panel of Figure 9. Zeroing the visual entries at only the three earliest attention layers reproduces 15.8 of the 17.2 points lost when all thirteen are zeroed. Zeroing the three latest layers costs only 0.5 points. So visual information enters through the early attention layers and is consumed downstream. (For these per-layer conditions we zero entries rather than delete them, because Zamba2’s attention blocks are shared and accept a single mask length. Where both operations are available, at $d=32$, they agree to within 2.34 points, which validates the substitution.)

Finally, the contrast is stable across the choice of projection basis. Repeating it with three independent Gaussian projections gives, for the fast tercile at 32 tokens, intact $31.2 \pm 1.6\%$ against evicted $9.9 \pm 1.8\%$ (mean and standard deviation over seeds), while the slow tercile shows no eviction effect at any seed.

8.2 The decay parameters predict how much is relayed

Result 2 found that the parameter-derived horizon does not predict the measured decodability curve. It does predict something else, and that something is exactly what the relay account says it should predict. Reason it through: a head that forgets quickly must depend more heavily on being refreshed, so deleting the source of the refresh should hurt its readout more. Ordering the strata by decay rate shows exactly this pattern. The size of the effect depends on how the strata are formed:

split	stratum	predicted $t_{1/2}$	intact at $d=0$	drop under eviction
tercile	fast	1.98	79.7%	37.5 points
tercile	slow	28.96	62.5%	7.8 points
gain-matched	fast	2.30	52.6%	22.4 points
gain-matched	slow	16.35	69.8%	9.4 points

Table 6: Dependence on the attention channel at $d=0$, against the decay rate implied by each stratum’s own A and gate. Terciles: $n = 64$ per cell; gain-matched: $n = 192$ per cell. The tercile contrast reaches $4.8\times$ but confounds decay with write amplitude; at matched amplitude the ratio is $2.4\times$. The two splits come from different collections (the gain-matched one carries a long pre-image passage to support the matched text eviction of Section 8.1), so accuracies are comparable within a split, not across splits.

Note that each row of this table compares intact against evicted *on the same heads*, so write amplitude cancels within a row. The amplitude confound affects only comparisons between strata. This is why the ratio worth quoting is the gain-matched one. At matched write gain, the fast stratum loses 22.4 points beside the image where the slow stratum loses 9.4, a ratio of $2.4\times$ at equal amplitude. At thirty-two tokens the gain-matched fast stratum reads only five points above chance even when intact, so eviction has little left to remove, and the contrast is uninformative there.

So the decay calculation does carry predictive content, but about the mechanism rather than about the decay curve: it identifies which heads are relaying and which are retaining. The effect is half the size the confounded split suggested, and it is measured at one distance on two strata. We flag both points in the limitations.

The headline contrast is large. At 32 tokens, the fast tercile reads 51.6% with the attention channel intact and 7.8% without it, which is below the 12.5% chance rate. In raw counts, thirty-three correct predictions out of sixty-four become five. Meanwhile the slow heads at distance zero, whose content was written while the image was still being read, are barely affected by eviction (62.5% to 54.7%). This is the control showing that the intervention does not simply break the model.

What a probe reads out of the recurrent state at distance is not what the state remembered. It is what the attention channel is still supplying, continuously re-supplied while that copy is present. Our evidence does not distinguish re-encoding at every token from a single re-supply shortly after the image, so we claim the dependence, not its period. The recurrent channel relays the picture; it does not store it.

8.3 What the probe can see, and why that matters here

There is one more objection to consider: perhaps the probe is simply insensitive to weak signals. Freshly re-encoded content plausibly sits at higher amplitude than content written long ago and decayed. A probe that finds strong signals and misses faint ones could then produce all of our observations even if the state never stopped storing the image.

This objection can be measured, and we measured it. We take the distance-zero features, where the image is present at full strength, and artificially weaken each trial’s own contribution while substituting an equal amount of unrelated variation from another trial, keeping total variance roughly constant: $x_i(\alpha) = \mu + \alpha(x_i - \mu) + \sqrt{1 - \alpha^2}(x_j - \mu)$. This transformation mimics what decay followed by overwriting does to a stored trace: the signal shrinks while unrelated content takes its place.

The result gives the probe a detection floor. The probe loses half its signal at $\alpha = 0.78$ and cannot be distinguished from chance by $\alpha \approx 0.25$ (Figure 10). The best single-layer probe behaves the same way, halving at $\alpha = 0.75$. In plain terms, this instrument cannot see content below roughly a quarter of full amplitude.

Now put that floor next to the prediction. The weights say that after thirty-two tokens, a retained trace sits at 3.7% of full amplitude at the median head and 18% at the mean, both below the floor. So a genuinely retained trace at that distance *could not* produce a visible reading. Yet the intact probe reads the fast heads at 51.6% there. Whatever the probe is reading must therefore be high in amplitude, which means recently written. And deleting the attention channel’s copy removes it.

The probe’s insensitivity to faint signals is what makes the positive reading informative. A signal the instrument can see at thirty-two tokens is necessarily a fresh one, because a retained one would be four to six times below its detection floor.

The same measurement bounds what we may conclude. Whether faint retained content persists below the floor is not answerable with this instrument at any distance beyond a few tokens, so this paper does not claim the state stops storing the image. It claims that what a probe reads at distance is not retention, and that it depends causally on the attention channel.

This one measurement explains three earlier puzzles. First, it explains why the measured horizon exceeds the predicted one, and why the gap grows in smaller models: the probe was never measuring retention in the first place. Second, it explains why the tercile split made the fast heads look like the best retainers: they write hardest, so whatever the attention channel re-supplies lands in them at the highest amplitude, and holding the write gain fixed removes that apparent advantage (Section 7.3). Third, it supplies the mechanism behind the causal result, which the next section takes up.

9 Result 4: Causally, the Answer Follows the Attention Channel

The probe results say the attention channel supplies what the recurrent state appears to hold. The splice tests the same claim on the model’s own behaviour, with no probe involved. Recall the setup: two runs see different images, one memory channel is exchanged between them, and the host run then answers its question. We admit a pair only if the host and the donor each answered their own question correctly beforehand, so the ceiling is 100% by construction. Of 111 attempted pairs, 11 were rejected on that criterion, and none were rejected for mismatched cache length. Every recurrent swap was verified to have actually changed the state, 100 of 100.

Figure 11 draws the same result as a curve, and three things can be read from it. First, replacing the recurrent memory has no effect at all: exchanging all 81 recurrent states and convolution buffers leaves 100 of 100 answers following the host’s own image, with every swap verified to have taken effect. Second, the attention channel is causally sufficient to

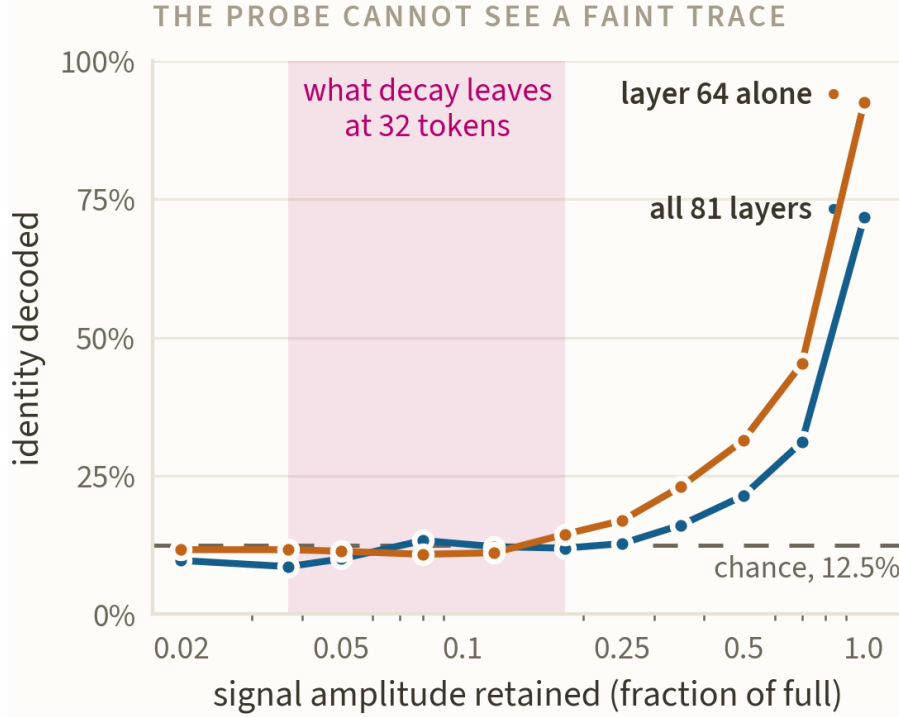


Figure 10: How faint a signal the probe can still recover. Each trial’s own contribution is attenuated while an equal amount of unrelated variation is substituted, holding total variance roughly constant. The shaded band is what the model’s decay parameters leave of a stored trace after 32 tokens, between the median and the mean over heads. Both probes are at chance throughout that band, so a visible reading at 32 tokens cannot be a retained one.

condition	layers swapped	follows host	follows donor
no swap	0	100/100 [96, 100]	0/100 [0, 4]
recurrent, all 81 states	0	100/100 [96, 100]	0/100 [0, 4]
attention, 1 layer	1	100/100 [96, 100]	0/100 [0, 4]
attention, 3 layers spread	3	98/100 [93, 99]	0/100 [0, 4]
attention, 7 layers	7	4/100 [2, 10]	94/100 [88, 97]
attention, all 13	13	0/100 [0, 4]	98/100 [93, 99]
attention, earliest 3	3	94/100 [88, 97]	1/100 [0, 5]
attention, latest 3	3	100/100 [96, 100]	0/100 [0, 4]
both channels	13	0/100 [0, 4]	100/100 [96, 100]

Table 7: The splice at 100 pairs with a 100% ceiling. Brackets are 95% Wilson intervals in percent. Replacing the entire recurrent memory of all 81 layers changes nothing; replacing seven of the thirteen attention caches flips the answer to the other image.

determine the answer: swapping seven of the thirteen caches moves 94 of 100 answers to the donor’s image. Third, the dependence on attention is graded and distributed. One swapped attention layer changes nothing, three spread across the stack move two answers, and seven move almost all of them. No small subset of layers is enough, so the picture is held across the attention layers rather than concentrated in a few. The earliest three layers have a small effect and the latest three have none, which matches the earlier finding that visual information enters early and is consumed downstream.

Why does replacing 37 million recurrent numbers change nothing? Result 3 already answers this. The transplanted state is decayed away within a few tokens and rewritten from the attention channel, which the recurrent swap leaves untouched. The probe result and the splice result are the same fact seen from two sides.

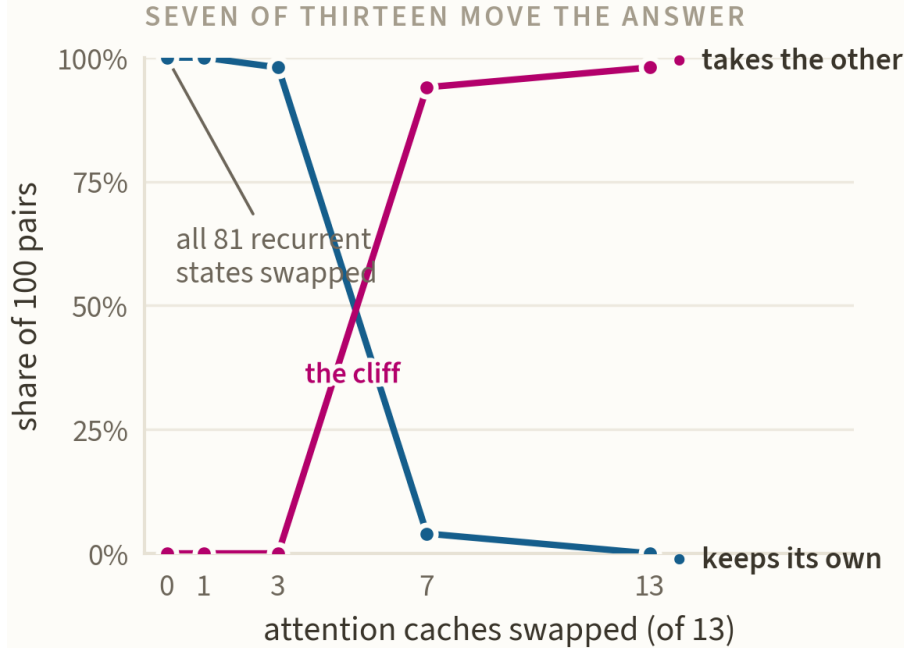


Figure 11: How many of the thirteen attention caches must be exchanged before the answer follows the other image, over 100 pairs in which both images were answered correctly beforehand. The leftmost point is the recurrent swap: all 81 states replaced, no attention cache touched, no effect. Then notice the cliff: one swapped cache changes nothing, three change almost nothing, and seven flip nearly every answer.

10 Result 5: Scale, and Where the Prediction Fails

We repeated the core protocol at three model scales. The instruments read every tensor shape from the loaded model’s configuration rather than assuming the 7B geometry. This is a requirement, not a convenience: the 2.7B and 1.2B checkpoints use a single B/C group where the 7B uses two, and the 1.2B has a state dimension of 128 where the others have 64. Table 8 reports the results.

model	layers	attn	probe $d=0$	probe $d=64$	measured $t_{1/2}$	predicted $t_{1/2}$
Zamba2-VL-7B	81	13	71.7%	21.7%	4.0	6.64
Zamba2-VL-2.7B	54	9	84.2%	31.7%	9.6	2.91
Zamba2-VL-1.2B	38	6	83.3%	34.2%	14.3	2.58

Table 8: The same protocol at three scales, 120 identity trials per distance. The pre-image control reads exactly 12.5% at every scale. Half-lives are in tokens: measured is the distance at which above-chance decodability halves, predicted is the median over heads of $\ln 2/(|A|\bar{\Delta})$.

Two conclusions follow. The second is a negative result, and we report it plainly.

First, the parameter-derived horizon is short at every scale. Median predicted half-lives are 6.64, 2.91 and 2.58 tokens, and the fraction of heads with a predicted half-life under 32 tokens is 85.0%, 97.9% and 95.6%. Whatever the recurrent state genuinely retains of its own past, it retains it for single-digit token counts in all three models.

Second, the predicted horizon does not predict the measured one. Across the three models the two quantities move in opposite directions: the prediction falls with scale while the measurement rises. On the 7B alone the agreement looked strong, with measured decodability tracking predicted retention at $r = 0.973$ across eleven distances. That agreement does not replicate at the other scales. We therefore do not claim that the decay calculation explains the measured decay curve. Result 3 already showed why it cannot: the probe at distance is reading relayed content, not retained content, so its curve is not governed by the decay parameters.

11 Result 6: Visual Key-Value Eviction Removes the Only Copy, Eventually

An image adds hundreds of key-value entries to the attention cache, which makes vision-language models expensive to serve. Pruning methods make them cheaper by deleting most of those entries, on the premise that the entries are redundant: on standard benchmarks, many can be deleted and accuracy barely moves. Our results explain where that redundancy comes from. It comes from the relay, and the relay only lasts a few tokens, so the premise should degrade with distance. We test this directly on the model’s own behaviour. We evict a fraction of the visual key-value entries from all 13 attention layers immediately after the image is read, let the filler and the question stream past the pruned cache, and measure the model’s accuracy. Retention ρ is the fraction of visual positions kept, sampled uniformly, and the question is asked d tokens after the image.

visual KV retained	$d=0$	$d=32$	$d=128$	$d=512$
100%	97.9%	95.8%	100.0%	97.9%
25%	91.7%	95.8%	95.8%	91.7%
5%	87.5%	77.1%	75.0%	64.6%
0%	64.6%	37.5%	31.2%	22.9%

Table 9: Behavioural accuracy under visual key-value eviction, against distance between the image and the question. $n = 48$ per cell, chance 12.5%, no failures. The unpruned row is flat, so distance alone costs nothing.

Read as cost relative to the unpruned baseline at the same distance, the structure is a clean interaction:

visual KV retained	$d=0$	$d=32$	$d=128$	$d=512$
25%	6.2	0.0	4.2	6.2
5%	10.4	18.8	25.0	33.3
0%	33.3	58.3	68.8	75.0

Table 10: Accuracy points lost to pruning, relative to the unpruned baseline at the same distance.

Figure 12 plots the four retention levels against distance. With a quarter of the entries retained, there is no distance effect at all: the cost is 6.2, 0.0, 4.2 and 6.2 points, flat within noise. At that level the redundancy premise holds on this architecture, and we found no evidence against it.

Below that level, distance starts to matter, and it matters a great deal. At a twentieth retained, the cost triples across the range, from 10.4 points beside the image to 33.3 points at 512 tokens. With every visual entry removed, it runs from 33.3 to 75.0. Compare the unpruned row, which is flat across the same distances. So the problem is not that long prompts are hard. The problem is that long prompts become hard *once the visual entries are gone*, because by then the recurrent channel has nothing left to relay from.

The cost of heavy pruning is not one number. Above a retention threshold, somewhere between a quarter and a twentieth on this model, pruning costs the same at every distance. Below it, the cost grows with how far the question sits from the image: a method validated at short range will understate its cost at long range by a factor of three at 5% retention.

The zero-retention row also gives us the cleanest available measure of what the recurrent channel contributes on its own. With no visual entries anywhere in the attention caches, the model can only be answering from its recurrent states. Its accuracy falls from 64.6% beside the image to 22.9% at 512 tokens. That decline is the recurrent channel’s own contribution decaying, measured on behaviour, with an instrument entirely independent of the probe. It agrees with the picture the probe gives.

One discrepancy should be stated plainly. At zero retention the model still answers at 37.5% at thirty-two tokens, well above the 12.5% chance rate, while our linear probe reads the recurrent state at close to chance at that distance. So the model is using something in the recurrent channel that the probe does not recover. This means the probe is a lower bound on what the state carries, not a full measure of it. The sentence “the state is empty” is not supported by our evidence, and we do not claim it.

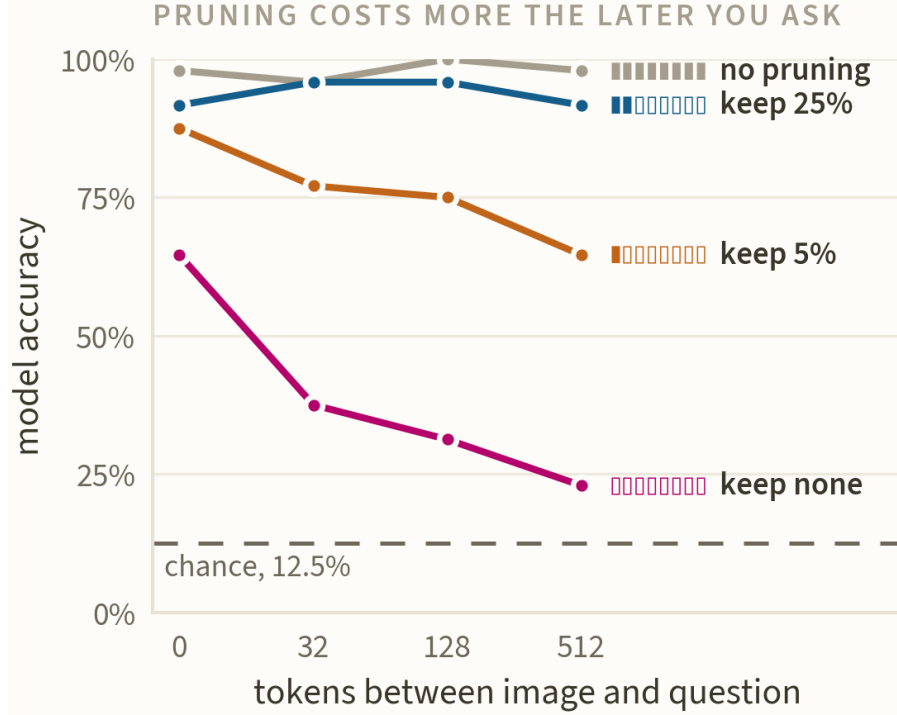


Figure 12: Model accuracy under visual key-value eviction, against how far the question sits from the image, $n = 48$ per point. The unpruned line is flat, so distance alone costs nothing. Keeping a quarter of the entries is also flat. Below that the lines fan out: the cost of pruning grows with distance only once the recurrent channel has stopped relaying the picture.

12 Limitations

We separate three kinds of limitation: alternative explanations our evidence does not exclude, limits on what the instruments can see, and limits on scope. The first kind is the one that should decide how much weight a reader puts on the central claim, so it comes first.

12.1 Alternative explanations we have not excluded

Eviction changes more than the image’s presence, though the matched control argues against this. Removing the visual key-value entries shortens every attention layer’s cache by roughly 400 positions, and it changes the softmax normalisation for every token that follows. Section 8.1 reports the matched control: deleting an equal number of *non-visual* positions from the same caches. The two deletions move the readout in opposite directions, which pure damage cannot produce. A narrower version of the worry survives. The pre-image text we remove does not contribute to the residual stream in the same way the image does, so the comparison bounds the damage account rather than eliminating every version of it.

The relay result could be a sensitivity asymmetry. We bound this but do not eliminate it. Our own data show the probe is a lower bound (see below), and a lower bound need not be uniform. Freshly re-encoded content plausibly sits at higher amplitude than content written many tokens earlier and decayed. If the probe recovers strong signals easily and weak signals poorly, removing the re-encoded copy could collapse the readout even in a state that still retained something. Section 8.3 answers this by measuring the probe’s detection floor and showing that predicted retention at thirty-two tokens sits below that floor, so a visible reading there cannot be a retained one. This argument makes one assumption: that decay followed by overwriting weakens the discriminative signal roughly the way our attenuation procedure does. A model with no attention channel to relay from would settle the question without that assumption, and we have not tested one.

The route is located but not traced. The layerwise eviction of Section 8.1 shows the relay enters early: zeroing the three earliest attention layers reproduces 15.8 of the 17.2 points lost when all thirteen are zeroed, while the three latest produce 0.5. What is not traced is the intermediate path, from attention output through the residual stream into a downstream mixer’s write.

12.2 What the instruments can and cannot see

The probe is a lower bound, and our own behavioural data prove it. With every visual key-value entry removed, the model answers at 37.5% at thirty-two tokens while the probe reads the recurrent state near chance at the same distance. Something usable survives there that a 512-dimensional random projection, read by a linear probe trained on 120 examples, does not recover. Every statement in this paper about what the state contains should be read as a statement about what is linearly decodable under that instrument. This paper does not establish that the state is empty of the image at any distance, and makes no such claim.

Probe accuracy depends on the feature-to-trial ratio. Count read at chance with 128 projection features and reached 19.4% at 512. In the other direction, the full 41,472-dimensional state probe reads 71.7% where a single layer’s 512 features read 92.5% on identical trials. Absolute accuracies are therefore not comparable across collections of different size or width, and we report them only within a collection. The decay curve in Table 2 is internally consistent because the same 120 trials appear at every distance.

One projection basis for most results. The eviction contrast was repeated across three independent Gaussian projection bases and is stable under them (fast tercile at $d=32$: intact $31.2 \pm 1.6\%$ against evicted $9.9 \pm 1.8\%$, mean and standard deviation over seeds). The decay-curve collections still use a single fixed basis, so the variance attributable to the choice of basis is unmeasured there.

The stratum dose-response is confirmed at matched amplitude only at distance zero, in one model. On splits by half-life the eviction sensitivity is ordered exactly by predicted decay rate: 37.5, 18.8 and 7.8 points over terciles ($r = 0.991$), and $r = 0.800$ (95% CI [0.34, 0.95]) at 32 tokens over deciles with about 6 points of binomial noise per cell at $n = 64$. But every split by half-life is also a split by write amplitude, and the amplitude-controlled version is thinner: a $2.4\times$ ratio at distance zero over two strata at $n = 192$, and no measurement at thirty-two tokens, where the gain-matched fast stratum reads too close to chance intact for eviction to have anything to remove. Two strata at one distance in one model is a direction, not a curve.

12.3 Scope

One model family, and a second family we tried and failed on. Results come from three Zamba2-VL checkpoints, which is one vendor, one architecture family and one training recipe. This is the principal limit on the paper’s generality.

We attempted to extend the relay result to a second family and report the attempt because its failure is informative. Granite 4.0 h-tiny [36] is a Mamba2 hybrid from a different lineage, with 36 recurrent layers to 4 attention layers against Zamba2-VL-7B’s 81 to 13. Its decay parameters read out cleanly and give a median predicted half-life of 8.7 tokens, so the horizon calculation of Section 5 transfers to a second family. The relay contrast does not, for an instrumental reason: on a code-word recall task the model answers correctly on 100%, 96.9% and 100% of trials at 0, 32 and 64 tokens, and yet a probe on its recurrent state reads at chance at distance zero. Scaling to 256 trials did not change this (13.7% against a 12.5% chance rate, $p = 0.31$), and neither did probing single layers, where the best of 36 layers reached 19.9%.

We suspected the item was too small to write a detectable trace, a one-token code word against a 345-token image, and tested that by repeating the carrier sixteen times so the item occupied 192 tokens. It did not help: 12.0%, 17.2% and 10.9% across the three strata. Repetition lengthens an item without diversifying what it writes, so the state ends up holding little more than the final repeat, which is not the manipulation we intended. We do not know why the instrument fails on this model; we only know that it does. No relay claim can be made beyond the Zamba2-VL family on this evidence. A vision-language hybrid from a second family, where the item is a real image, remains the right test and we have not run it. The sharper version of that test is a vision-language model with no attention layers at all, such as Cobra [32] or VL-Mamba [33], where probing the recurrent state must be measuring retention because there is nothing else it could be measuring. Both are Mamba1 models and would need the read operator ported before any number from them could be trusted.

One task format, one dataset, one modality. Eight-way forced choice on COCO buys an exact chance rate and a clean pre-image control, and costs generality to free-form answering, to other visual properties, and to whether any of this holds for text held in the same states.

The horizon is measured on one filler distribution. $\bar{\Delta}$ is input dependent. We measured it on natural prose and found the median half-life stable across images, from 6.44 to 6.86 tokens, but a filler with very different gate statistics would give a different horizon, which the same calculation would predict.

13 Conclusion

The recurrent state of this hybrid vision-language model is not a memory of the image in any useful sense. It is a selective summary of the picture’s gist, and it holds that summary for only a few tokens. This lifetime is not an empirical curiosity: it can be computed from the model’s own decay parameters before any measurement is taken. What the model actually answers from lives in the thin attention channel, and the recurrent state relays that content rather than storing its own copy. The practical consequence is direct: the visual key-value entries a hybrid carries are redundant only for as long as the recurrent state is still relaying the picture, and that is not long.

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